Question 2: Innovation is the key to prosperity in an age of new technologies

Extract 5: United Kingdom (UK) government to bring forward ban on new petrol and diesel cars

Emission of nitrogen oxides (NO_x) from car exhausts regularly exceeds safe levels. Studies have linked it to heart and lung diseases as well as other conditions. Petrol and diesel vehicles produce the overwhelming majority of roadside NO_x gases. The removal of petrol and diesel vehicles from the roads should help bring air quality within the guidelines set out by the World Health Organisation (WHO).

The UK government is to bring forward a ban on new petrol and diesel cars to 2035, five years earlier than previously planned. About a third of NO_x emissions in the UK come from transport. The ban on conventional car sales will help move the UK towards the net-zero emissions target that the government announced in 2019. The Committee on Climate Change (CCC), which advises the UK government, said if other countries followed the UK, there was a good chance of staying below the recommended maximum global temperature rise of 1.5 degrees Celsius (°C) by 2100. This is considered the threshold to avoid dangerous climate change. However, banning new petrol and diesel cars will not be enough on its own. The government will also need to address the emissions coming from domestic and industrial energy use.

The CCC believes the price of electric cars will be similar to that of petrol or diesel cars by 2024–25. But for now, electric cars are more expensive than their petrol or diesel substitutes. This is because battery technology is still developing and production requires expensive raw materials. The UK government currently provides a £3500 subsidy for the purchase of a new electric car, but for many consumers, the price is still too high.

Some experts say a more radical approach, such as no sales tax on electric cars, would create a large increase in demand. A scheme like this exists in Norway, where there was a significant increase in the sales of electric cars in 2019.

Source: BBC News, 4 February 2020

Extract 6: The German car industry faces challenges

A crisis is sweeping the German car industry. Two leading German car producers announced more than 20 000 job losses in the first signs of the human cost of the sector's transition from petrol and diesel driven cars to electric cars. Volkswagen is seeking to become a world leader in battery-powered cars. It is estimated that the German car industry, which directly employs 830 000 people and supports a further 2 million in the wider economy, will be forced to invest some 40 billion euros (€40billion) into battery-powered technologies over the next three years. German car manufacturers and component suppliers have announced that around 50 000 jobs will be lost this year, as their traditional businesses become less profitable. The global economic slowdown and the US–China trade war have forced manufacturers to revise their sales projections downwards. They had expected to sell in excess of 100 million vehicles in 2019. With just a few weeks left in the year, that figure is likely to be closer to 90 million.

Increasingly localised production by car makers in China and the US to avoid tariffs has also reduced the exporting power of German-based firms, which are left dependent on the more fragile European market. To make matters worse, demographic change could permanently reduce the number of new car buyers. The German car industry needs to adapt and change faster. In the next decade, almost a quarter of a million jobs in car production will be lost in the country. Car sales in China, which helped Germany's biggest brands survive the financial crisis, have slowed for 17 months in a row, drastically reducing a key source of revenue at the precise moment it is needed to fund new technologies.

Strict European Union (EU) carbon-emissions rules are forcing car makers to accelerate their plans to switch to producing electric vehicles, or face billions of euros in fines. The EU's targets for 2030 will mean that between 7 million and 10.5 million battery-powered cars will be on Germany's roads by the end of the next decade. Volkswagen's first mass-market battery-powered family car is rolling off production lines in Germany.

The cost of job losses in Germany, estimated at €100 000 per job, forces companies to consider reskilling programmes or wait for employees to retire. Yet even the German car industry predicts that the shift to electric cars will cause the loss of 70 000 jobs in the near future, as their assembly is far less labour-intensive.

Industry insiders claim that there is still reason to be optimistic. The engineering expertise that has helped Germany's car makers survive for 130 years is expected to carry them through the wave of disruption that looms on the horizon. The German car industry is expected to become a pioneer in electric car production.

Source: Financial Times, *30 November 2019*

Extract 7: Bloomberg Innovation Index: Germany replaces South Korea as most innovative nation

Table 1: 2020 Bloomberg Innovation Index (top 12 of 50 economies shown)

Economy	2020 rank	2019 rank	R&D intensity	Productivity	Tertiary education efficiency
Germany	**1**	2	8	18	26
S. Korea	**2**	1	2	29	16
Singapore	**3**	6	12	4	1
Switzerland	**4**	4	3	14	17
Sweden	**5**	7	4	19	13
Israel	**6**	5	1	15	32
Finland	**7**	3	10	9	24
Denmark	**8**	11	7	6	31
US	**9**	8	9	12	47
France	**10**	10	13	16	20
Austria	**11**	12	6	13	12
Japan	**12**	9	5	35	30

In economic terms, innovation describes the development and application of ideas and technologies that improve goods and services or make their production more efficient. Innovation is an essential driver of economic progress that benefits consumers, producers and the economy as a whole. It has two main aspects. Product innovation involves the introduction or improvement of goods or services. Process innovation involves the introduction of improved production techniques.

The Bloomberg Innovation Index uses seven criteria to rank 50 economies. These criteria include research and development (R&D) spending, productivity and tertiary efficiency.

In 2020 Germany was ranked in first place after a period of six years when South Korea was ranked first. The German manufacturing sector is still highly competitive and a source of innovation. However, there are several reasons why Germany should not be complacent about its overall innovation ranking. Innovation in the German services sector is much less impressive and about a third of R&D spending is in the car industry. This means that if the car industry is disrupted it could decrease Germany's overall strength and innovation performance. Lack of innovation in the services sector in Germany is an increasing worry, especially as the global economy shifts more towards services and away from manufacturing. The German government would be well advised to use its on-going fiscal surplus to invest and safeguard Germany's role as an innovator in a wider range of economic sectors.

Singapore has risen in the index to third place overall. It was in sixth place last year, but has experienced productivity and manufacturing gains, while it retains a best-in-world ranking for tertiary-education efficiency. Innovation is a critical driver of growth and prosperity. China has moved up the rankings, but the US's fall in the index illustrates that trade tariffs are not going to maintain its economic advantage if there is a lack of investment in education and research that can lead to innovation.

Source: Michelle Jamrisko, Wei Lu, Bloomberg, 18 January 2020

Extract 8: Creating an innovation culture – Singapore's formula to becoming a regional tech hub

For some years the Singapore government has put in place policies designed to make itself into an innovation and start-up hub in the region. Singapore has also become a magnet for foreign technology firms, hosting regional headquarters and engineering hubs for many of the world's leading companies. A major British electronics company said earlier this year that it would be moving part of its operations to Singapore. In addition, several Chinese companies also have R&D centres in the country.

For a small country with few natural resources and a population of just 5.6 million, how did Singapore transform itself into a leading centre for technology and innovation? It comes down to creating what a Singapore minister called 'enabling conditions', such as policies that are friendly to businesses, a coordinated approach between universities and private companies to provide a trained work-force, and a high quality of life to attract globally-mobile entrepreneurs and top-grade talent who can call anywhere home. The country is also helped by the continuity of government policies and a highly educated local workforce. Singapore's status as a financial centre has also made it easy to attract those with capital looking to invest in Southeast Asia. Investments in advanced technology sectors have made it a base for high-end manufacturing, capable of developing electric cars, for instance.

The Singapore government has invested S$19 billion to build the country into a global R&D hub. In March, an additional S$500 million was set aside to expand investments in a range of new technologies. The Singapore government also takes the protection of intellectual property seriously, making it an attractive place for companies to undertake R&D.

Although it has an abundance of capital, Singapore has faced talent shortages, especially with a small population compared with major economies like the US and China. Attracting foreign talent is one solution, but Singapore has also put in place schemes to train locals in the skills needed for its technological goals.

Source: Zen Soo and Chua Kong Ho, South China Morning Post, *7 September 2019*

(a) Distinguish between a rise in total output and a rise in productivity. [2]

(b) Explain **one** reason why a firm with significant market power and long-run excess profits might choose to spend large sums on research and development (R&D). [2]

(c) Explain, with reference to Extract 8, what might be the opportunity cost of the S$19 billion invested by the Singapore government to build the country into a global R&D hub. [2]

(d) Explain how the creation of an innovation culture in Singapore is likely to have benefited the Singapore economy. [6]

(e) Discuss whether subsidising the purchase of electric cars would improve the efficiency of resource allocation in the market for transport in Singapore. [8]

(f) Discuss the extent to which government policy can influence a country's comparative advantage in a good or service. [10]

[Total: 30]